iLearn Practice Exam

12BR · Chapter / Topic 20

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Topic 20: Further Programming

Name: _____

Class: _____

Date: _____

Time allowed: 45 minutes

TOTAL MARKS: 37

Marks obtained: ..... / 37

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 37.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 5 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 37 marks of past-paper questions are available for this chapter, so the recommended time is 45 minutes (1 min 1

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1 **(a)** A medical centre uses objects of the class `Appointment` to record treatments given and medication prescribed during each doctor's appointment. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
`DateSeen`	`DATE`	date of treatment
`Treatments`	`STRING`	treatments given
`Medications`	`STRING`	medications prescribed

Patients are identified by a unique 8-digit number, beginning with the patient's year of birth, for example, 20108989.
Doctors are identified by their name, for example, A N Other.

Complete the class diagram for `Appointment`, to include:

- attribute and data type for the identification of the patient
- attribute and data type for the identification of the doctor
- methods to assign date seen, treatments given and medications prescribed
- method to return the date seen and the attributes for the patient and the doctor.

Appointment
DateSeen : DATE ..... : .... : ..... Treatments : STRING Medications : STRING
..... SetPatientID(PatientNumber : INTEGER) SetDoctor(DoctorID : STRING) .... GetTreatments() GetMedications()

[5]

(b) (i) Identify the object-oriented programming (OOP) feature whose function includes restricting external access to the data.

..... [1]

(ii) Describe what is meant by the OOP feature **inheritance**.

.....

.....

.....

..... [2]

2 **(a)** Object-Oriented Programming (OOP) is used to design a class called `Node`.

A partial design of the `Node` class includes the `STRING` attribute `Content` and the `INTEGER` attribute `Pointer`, together with their getters and setters.

The class also includes a constructor, with two parameters, to set the values of these attributes.

Write pseudocode statements to declare the `Node` class, its attributes and the constructor.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) Complete the table by filling in the missing OOP terms and descriptions.

OOP term	description
.....	A method used to change values in private attributes in a class.
.....	An occurrence of a class during the execution of a program.
aggregation	....
inheritance	....

[4]

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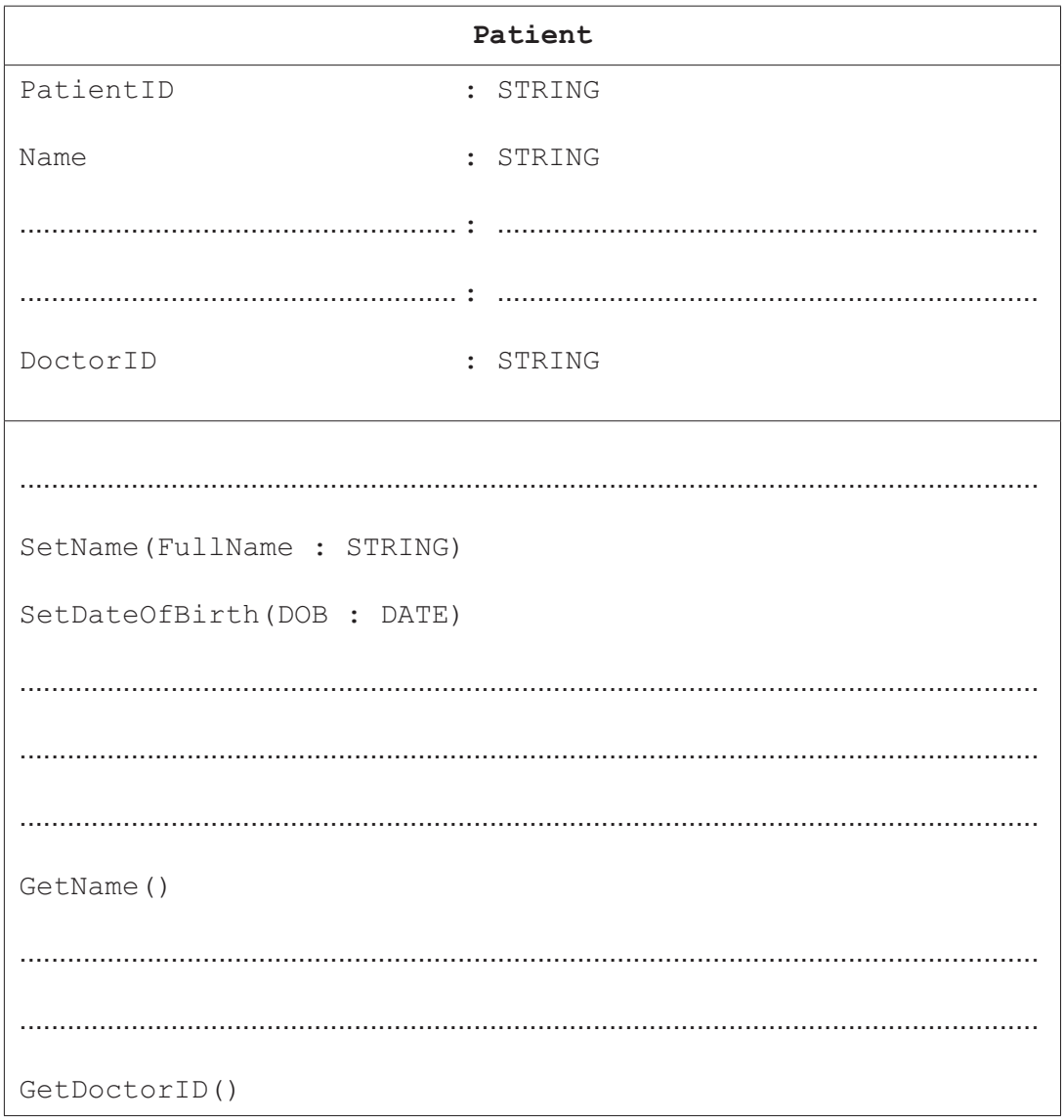
3 **(a)** A medical clinic uses objects of the class `Patient` to assign a priority and a doctor to a patient. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
`PatientID`	`STRING`	Unique identifier of the patient
`Name`	`STRING`	Patient's full name, surname first
`DoctorID`	`STRING`	ID of doctor administering treatment

Treatment is prioritised with a numeric scale of 1 to 5.

Complete the class diagram for `Patient`, to include:

- attribute and data type for the date of birth
- attribute and data type for the priority
- methods to assign the patient ID, priority and doctor ID
- methods to return the patient ID, patient date of birth and the priority.



[5]

(b) **(i)** Identify the object-oriented programming (OOP) term described as ‘an occurrence of an object’.

..... [1]

(ii) Describe what is meant by the OOP term **polymorphism**.

.....

.....

.....

..... [2]

4 A declarative programming language is used to show which home cinema devices customers already have and which devices they want.

```
01 device(projector).
02 device(large_tv).
03 device(medium_tv).
04 device(bluray_player).
05 device(amplifier).
06 device(media_player).
07 device(main_speakers).
08 device(centre_speaker).
09 device(surround_speakers).
10 customer(petrov).
11 customer(margaret).
12 customer(steffan).
13 customer(susan).
14 customer(allan).
15 customer(sammy).
16 owns(projector, petrov).
17 owns(media_player, petrov).
18 owns(projector, susan).
19 owns(projector, sammy).
20 owns(media_player, allan).
21 owns(bluray_player, steffan).
22 wants(main_speakers, margaret).
23 wants(surround_speakers, susan).
24 wants(medium_tv, steffan).
```

These facts (clauses) have the meanings:

fact	meaning
01	A projector is a device.
10	Petrov is a customer.
16	Petrov owns a projector.
22	Margaret wants main speakers.

(a) Betty is a customer who owns a large TV and wants a Blu-ray player.

Write additional facts to represent this information.

25 .....

26 .....

27 .....

[3]

(b) Using the variable `Owner`, the goal:

```
owns(media_player, Owner)
```

returns

```
Owner = petrov, allan
```

Write the result of using the goal:

```
owns(projector, Owner)
```

`Owner =` ..... [1]

(c) `D` may be available for `A` to buy if `D` is a device and `A` is a customer and `A` does **not** own that device.

Write this as a rule:

```
may_buy_device(D, A)
```

`IF` .....

.....

.....

..... [4]

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5 Complete the table by filling in the missing object-oriented programming (OOP) terms and descriptions.

OOP term	Description
.....	A method that accesses the value of a property.
.....	A method that changes the value of a property.
Object	....
Method	....

[4]

END OF QUESTION PAPER